Chamfer Distance as a Metric on Partitions

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1 Introduction

We introduce the *Partition Chamfer Metric* (PCh) as a new metric for comparing measurable partitions.

1.1 Chamfer Distance

The chamfer distance was introduced in 1977 by Barrow et al. [4] for image matching. Since its introduction, the chamfer distance has found use in the fields of computer vision and data science. For finite non-empty sets $X, Y \subset \mathbb{R}^n$ the typical definition of chamfer distance is given as:

$$CH(X,Y) = \frac{1}{|X|}\sum_{x\in X}\min_{y\in Y} d(x,y) + \frac{1}{|Y|}\sum_{y\in Y}\min_{x\in X} d(x,y)$$

where $d(x,y)$ is an L^p metric. Some variations include the directed chamfer distance:

$$\overrightarrow{CH}(X,Y) = \frac{1}{|X|}\sum_{x\in X}\min_{y\in Y} d(x,y)$$

and the unnormalized directed chamfer distance:

$$\overrightarrow{Ch}(X,Y) = \sum_{x\in X}\min_{y\in Y} d(x,y)$$

Despite its name, the chamfer distance is not a true metric as it fails to satisfy the triangle inequality. Consider the collection of three sets in $\mathbb{R}$

$$X = \{0\}, Y = \{1,5\}, Z = \{6\}$$

under the usual Euclidean metric as one such counterexample. Even so, the chamfer distance has often been used as a proxy for the earth mover distance as it is much faster to compute. A naive implementation of chamfer distance is quadratic in runtime and recent work [8] has found nearly linear algorithms for computing the chamfer distance. Since its introduction, variations such as the Density-Aware Chamfer Distance [9] have allowed for efficient computations of related "distances".

1.2 Steinhaus Distance

We will use the definition of the Steinhaus Metric given by Deza and Deza [7]. Consider a measure space $(\Omega, \mathcal{A}, \mu)$. Let $\mathcal{A}_\mu = \{a \in \mathcal{A} | \mu(a) < \infty\}$. Let $\bar{\mathcal{A}}_\mu$ be the set of equivalence classes of $\mathcal{A}_\mu$ where two sets $a, b \in \mathcal{A}_\mu$ are said to be equivalent if $\mu(a \triangle b) = 0$. The Steinhaus metric:

$$d_s(a, b) = 1 - \frac{\mu(a \cap b)}{\mu(a \cup b)}$$

(with the added requirement $d_s(a, b) = 0$ if $\mu(a) = \mu(b) = 0$) is a metric on $\bar{\mathcal{A}}_\mu$. This metric has found recent use in topological data analysis [2]. Furthermore, the notion of equivalence modulo sets of measure 0 will be extended in our main result.

1.3 Application of Partitions

Graph partitions have played a role in many fields including computing systems [3] and computational redistricting [6]. Within data science, the clustering problem is also an application of partitions. Comparing the similarity between two clusters often relies on Earth Mover's Distance (EMD) [1].

2 Acknowledgment of AI Use

We constructed the Partition Chamfer Metric (PCh) independent of LLMs when considering properties of partitions. For approximately a year, we attempted to prove the triangle inequality for this metric with and without LLM assistance to no avail. In particular we tried using GPT 5.2, GPT 5.3, GPT 5.4, and Copilot (the free model available in January 2026). We found all the details of the proof except for the similarity lemma without LLM use. We used Qwen 3.8 to prove the original form of this lemma in July 2026. After seeing Qwen's proof, we extended the result from finite partitions of finite Ω to countable partitions of (possibly) infinite cardinality Ω. We have taken care to verify the proof and tried our best to make the proof human understandable. The required mathematics background for the cleaned proof is typical of a first year math graduate student's coursework and care was taken to find and cite the originators of ideas when possible. This work is still in progress, but given recent LLM math results and practices from OpenAI, we opted to post our current results. These results came from working with open weight models and careful human interaction with LLMs.

3 Main Result

A list of the notation used throughout the proof is summarized in 1 below.

Table 1: Notation used in the Main Result section.

Symbol	Meaning
Measure space and Steinhaus metric	

Table 1 continued

Symbol	Meaning
$M = (\Omega, \mathcal{A}, \mu)$	measure space with $0 < \mu(\Omega) < \infty$
Ω	ground set (the whole region carrying the measure)
$\mathcal{A}$	σ-algebra of measurable subsets of Ω
μ	measure on $\mathcal{A}$
$a \triangle b$	symmetric difference $(a \setminus b) \cup (b \setminus a)$
$d_s(a, b)$	Steinhaus metric $1 - \mu(a \cap b)/\mu(a \cup b)$, with $d_s(a, b) = 0$ when $\mu(a) = \mu(b) = 0$ (defined in the Steinhaus Distance subsection)
$J_s(x, y)$	Steinhaus similarity $1 - d_s(x, y) = \mu(x \cap y)/\mu(x \cup y)$
Measurable partitions and the isomorphism relation	
X, Y, Z	measurable partitions: countable collections of nonempty pairwise disjoint measurable parts with $\bigcup_{x \in X} x = \Omega$
x, y, z	parts (elements) of the partitions X, Y, Z
$\tilde{X}$	positive-measure parts of X: $\{x \in X : \mu(x) > 0\}$
f, g	measure-partition isomorphisms: bijections $\tilde{X} \to \tilde{Y}$ with $\mu(x \triangle f(x)) = 0$ for all $x \in \tilde{X}$; X, Y are *isomorphic* when such a map exists
$h = g \circ f$	composition of two isomorphisms (transitivity step); distinct from the atom-label map $h(S)$ below
$\bar{X}$	collection of equivalence classes of measurable partitions on M under the isomorphism relation
Partition Chamfer Metric	
$PCh(\tilde{X}, \tilde{Y})$	Partition Chamfer Metric; an extended metric on $\bar{X}$
$\overrightarrow{PCh}(\tilde{X}, \tilde{Y})$	directed PCh, $\sum_{x \in \tilde{X}} \min_{y \in \tilde{Y}} d_s(x, y)$; $PCh(\tilde{X}, \tilde{Y}) = \overrightarrow{PCh}(\tilde{X}, \tilde{Y}) + \overrightarrow{PCh}(\tilde{Y}, \tilde{X})$
Similarity Lemma and its proof	
$\mathcal{F}$	finite pairwise disjoint collection of positive-measure sets (lemma hypothesis); in the main proof, a finite subset of $\tilde{A}$
b, c	positive-measure measurable sets (lemma); in the main proof, fixed elements of $\tilde{B}, \tilde{C}$
$\mathcal{R}$	positive-measure sets in the σ-algebra generated by $\mathcal{F} \cup \{b, c\}$
$\mathcal{R}'$	atoms of $\mathcal{R}$ (elements with no proper subsets in $\mathcal{R}$); r denotes an atom
T_r	independent exponential random variable with rate $\lambda_r = \mu(r)$
$T_1, \ldots, T_n$; $\lambda = \sum_{i=1}^n \lambda_i$	generic exponential variables and the rate of their minimum (standard fact)
S	union of atoms in $\mathcal{R}'$
$h(S)$	atom-label map $\operatorname{argmin}_{r \in \mathcal{R}', r \subseteq S} T_r$; distinct from $h = g \circ f$ above
E, G	generic elements of $\mathcal{R}$ (MinHash-type identity $\mathbb{P}(h(E) = h(G)) = J_s(E, G)$)
$\mathbb{P}, \mathbb{E}$	probability and expectation
$\mathbf{1}_{\{\cdot\}}$	indicator function of an event
$\alpha_a, \beta, \gamma_a$	indicators $\mathbf{1}_{\{h(a)=h(b)\}}$, $\mathbf{1}_{\{h(b)=h(c)\}}$, $\mathbf{1}_{\{h(a)=h(c)\}}$ for $a \in \mathcal{F}$
Proof of the triangle inequality	
A, B, C	the three partitions to which the triangle inequality is applied (generic here; A, B are reused as the concrete example partitions below)

Table 1 continued

Symbol	Meaning
a, b, c	parts of A, B, C; these also instantiate the lemma variables a, b, c
b_a	part of $\tilde{B}$ closest to a: $d_s(a, b_a) = \min_{b \in \tilde{B}} d_s(a, b)$
c_b	part of $\tilde{C}$ closest to b: $d_s(b, c_b) = \min_{c \in \tilde{C}} d_s(b, c)$
I_b	fiber $\{a \in \tilde{A} : b_a = b\}$; the nonempty I_b partition $\tilde{A}$
Example calculation	
A, B	the two example partitions of the region in Figures 2–4
A_i, B_j	parts of A, B ($i, j \in \{1, 2, 3, 4\}$), colored red, blue, orange, purple

3.1 Measurable Partitions

Consider a measure space $M = (\Omega, \mathcal{A}, \mu)$ such that $0 < \mu(\Omega) < \infty$. A measurable partition $X \subset \mathcal{A}$ is a countable collection of nonempty pairwise disjoint measurable subsets of Ω such that

$$\bigcup_{x \in X} x = \Omega$$

We will refer to the elements of X as the parts of X. We will let $\tilde{X} = \{x \in X | \mu(x) > 0\}$. Two measurable partitions, X and Y are isomorphic if there exists a bijection $f : \tilde{X} \to \tilde{Y}$ such that $\mu(x \triangle f(x)) = 0$ for all $x \in \tilde{X}$. We will call f a measure partition isomorphism (or isomorphism for short).

3.1.1 Proof of Equivalence Relation

To show this is an equivalence relation we will show reflexivity, symmetry, and transitivity.

Reflexivity If we set f to the identity map, reflexivity follows.

Symmetry Suppose X is isomorphic to Y. Then there exists an isomorphism $f : \tilde{X} \to \tilde{Y}$. As f is a bijection, we also have a bijection $f^{-1} : \tilde{Y} \to \tilde{X}$. Let $f(x) = y$ for convenience. Then

$$0 = \mu(x \triangle f(x)) = \mu(x \triangle y) = \mu\left(f^{-1}(y) \triangle y\right)$$

Transitivity Suppose X is isomorphic to Y and Y is isomorphic to Z with isomorphisms f and g respectively. Further, let $h = g \circ f$ be the composition of these isomorphisms. Then h is a bijection. For convenience, let $y = f(x)$ and $z = g(y)$. Then

$$\mu(x \triangle f(x)) = \mu(y \triangle g(y)) = 0$$

For sets x, y, z, we have $x \triangle z \subseteq (x \triangle y) \cup (y \triangle z)$ thus

$$\mu(x \triangle h(x)) = \mu(x \triangle z) \leq \mu(x \triangle y) + \mu(y \triangle z) = \mu(x \triangle f(x)) + \mu(y \triangle g(y)) = 0 + 0 = 0$$

So h is an isomorphism and thus X and Z are isomorphic.

3.2 Partition Chamfer Metric

Let $\bar{\mathcal{X}}$ be the collection of all equivalence classes of measurable partitions on M. The Partition Chamfer Metric (PCh)

$$PCh(\tilde{X}, \tilde{Y}) = \overrightarrow{PCh}(\tilde{X}, \tilde{Y}) + \overrightarrow{PCh}(\tilde{Y}, \tilde{X})$$

is an (extended) metric on $\bar{\mathcal{X}}$, where

$$\overrightarrow{PCh}(\tilde{X}, \tilde{Y}) = \sum_{x \in \tilde{X}} \min_{y \in \tilde{Y}} d_s(x, y)$$

Before we prove the triangle inequality, we will show the metric is well defined. We must show that the minimum inside the sum always exists. To do so, consider the Steinhaus similarity:

$$J_s(x, y) = 1 - d_s(x, y) = \frac{\mu(x \cap y)}{\mu(x \cup y)}$$

In particular, notice that for all $x \in \tilde{X}, y \in \tilde{Y}$ we have

$$J_s(x, y) = \frac{\mu(x \cap y)}{\mu(x \cup y)} \leq \frac{\mu(x \cap y)}{\mu(x)}$$

Since the parts of $\tilde{Y}$ are disjoint, measurable, and partition Ω up to a set of measure 0, we see

$$\sum_{y \in \tilde{Y}} \mu(x \cap y) = \mu(x \cap \bigcup_{y \in \tilde{Y}} y) = \mu(x)$$

So we see

$$\sum_{y \in \tilde{Y}} J_s(x, y) \leq \frac{\sum_{y \in \tilde{Y}} \mu(x \cap y)}{\mu(x)} = \frac{\mu(x)}{\mu(x)} = 1$$

As each $J_s(x, y) \geq 0$ for each y, then it follows that the series $\sum_{y \in \tilde{Y}} J_s(x, y)$ converges and thus there must be a $y \in \tilde{Y}$ such that $J_s(x, y)$ is maximized, and therefore this same y must minimize $d_s(x, y)$.

3.3 Proof of Triangle Inequality

Now we will show that PCh obeys the triangle inequality. The rest of the requirements for PCh to be a metric are straightforward to verify and are left for the reader.

Lemma 3.1. *Let $\mathcal{F}$ be a finite pairwise disjoint collection of measurable sets with positive measure and let b, c be measurable sets with positive measure. Then*

$$\sum_{a \in \mathcal{F}} J_s(a, c) \geq \sum_{a \in \mathcal{F}} J_s(a, b) + J_s(b, c) - 1 .$$

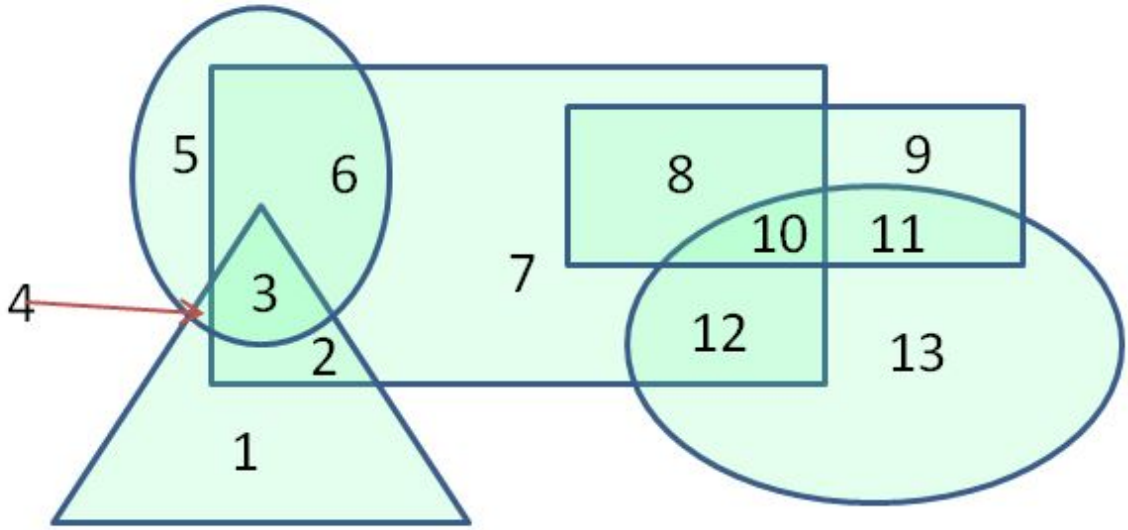


Figure 1: An Example of Atoms: If we use each outlined shape as a set and use area as our measure, then each numbered region is an atom.

Proof. Let $\mathcal{R}$ be the collection of all positive measure sets in the σ-algebra generated by $\mathcal{F} \cup \{b, c\}$. We will call any elements of $\mathcal{R}$ that have no proper subsets the atoms of $\mathcal{R}$. Let $\mathcal{R}'$ be the collection of atoms of $\mathcal{R}$. Note that these atoms are disjoint. An example of atoms is given in Figure 1.

For each $r \in \mathcal{R}'$, define an independent exponential random variable T_r with parameter $\lambda_r = \mu(r)$. Let S be a union of atoms in $\mathcal{R}'$. Then we define

$$\phi(S) = \underset{r \in \mathcal{R}', r \subseteq S}{\operatorname{argmin}} T_r$$

For exponential random variables, $T_1, T_2, \ldots, T_n$ with corresponding parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$ it is well known that $T = \min\{T_1, T_2, \ldots, T_n\}$ is exponentially distributed with parameter $\lambda = \sum_{i=1}^{n} \lambda_i$. Furthermore, it is well known for $1 \leq k \leq n$ that $\mathbb{P}(T_k = \min\{T_1, T_2, \ldots, T_n\}) = \frac{\lambda_k}{\lambda}$. This implies that for any two sets $E, G \in \mathcal{R}$, we have the following:

$$\mathbb{P}(\phi(E) = \phi(G)) = \frac{\sum\limits_{r \in \mathcal{R}', r \subseteq E \cap G} \lambda_r}{\sum\limits_{r \in \mathcal{R}', r \subseteq E \cup G} \lambda_r} = \frac{\sum\limits_{r \in \mathcal{R}', r \subseteq E \cap G} \mu(r)}{\sum\limits_{r \in \mathcal{R}', r \subseteq E \cup G} \mu(r)} = \frac{\mu(E \cap G)}{\mu(E \cup G)} = J_s(E, G)$$

The connection between probability and the Jaccard similarity for finite sets has been explored in the MinHash algorithm [5]. The above was an extension of this idea. Now define the following indicator functions

$$\alpha_a = \mathbf{1}_{\{h(a)=h(b)\}}$$

$$\beta = \mathbf{1}_{\{h(b)=h(c)\}}$$

$$\gamma_a = \mathbf{1}_{\{h(a)=h(c)\}}$$

Since the sets in $\mathcal{F}$ are pairwise disjoint, at most one of the a has $h(a) = h(b)$. As such, we see

$$\sum_{a\in\mathcal{F}} \alpha_a, \sum_{a\in\mathcal{F}} \gamma_a \in \{0,1\}$$

Now, we wish to prove the inequality below.

$$\sum_{a\in\mathcal{F}} \gamma_a \geq \sum_{a\in\mathcal{F}} \alpha_a + \beta - 1$$

We have three cases:
Case 1: $\sum_{a\in\mathcal{F}} \alpha_a = 0$
In this case, we see

$$\sum_{a\in\mathcal{F}} \gamma_a \geq 0 \geq 0 + \beta - 1$$

as $\beta \in \{0,1\}$.
Case 2: $\beta = 0$
In this case, we see

$$\sum_{a\in\mathcal{F}} \gamma_a \geq 0 \geq \sum_{a\in\mathcal{F}} \alpha_a + 0 - 1$$

as $\sum_{a\in\mathcal{F}} \alpha_a \in \{0,1\}$.
Case 3: $\sum_{a\in\mathcal{F}} \alpha_a = 1$ **and** $\beta = 1$
Then there exists $a \in \mathcal{F}$ such that $h(a) = h(b) = h(c)$ so

$$1 = \sum_{a\in\mathcal{F}} \gamma_a = \sum_{a\in\mathcal{F}} \alpha_a + \beta - 1 = 1 + 1 - 1 = 1$$

As a result of these cases, we see

$$\sum_{a\in\mathcal{F}} \gamma_a \geq \sum_{a\in\mathcal{F}} \alpha_a + \beta - 1$$

Now, we will apply linearity of expectation:

$$\mathbb{E}\left[\sum_{a\in\mathcal{F}} \gamma_a\right] = \sum_{a\in\mathcal{F}} \mathbb{E}[\gamma_a] = \sum_{a\in\mathcal{F}} \mathbb{P}(h(a) = h(c)) = \sum_{a\in\mathcal{F}} J_s(a,c)$$

Similarly,

$$\mathbb{E}\left[\sum_{a\in\mathcal{F}} \alpha_a\right] = \sum_{a\in\mathcal{F}} \mathbb{E}[\alpha_a] = \sum_{a\in\mathcal{F}} \mathbb{P}(h(a) = h(b)) = \sum_{a\in\mathcal{F}} J_s(a,b)$$

$$\mathbb{E}[\beta] = \mathbb{P}(h(b) = h(c)) = J_s(b,c)$$

So if we take expectation of both sides of the inequality, we see

$$\mathbb{E}\left[\sum_{a\in\mathcal{F}}\gamma_a\right] \geq \mathbb{E}\left[\sum_{a\in\mathcal{F}}\alpha_a + \beta - 1\right]$$

$$\sum_{a\in\mathcal{F}} J_s\left(a,c\right) \geq \sum_{a\in\mathcal{F}} J_s\left(a,b\right) + J_s\left(b,c\right) - 1\,. \qquad \square$$

3.3.1 Proof of Triangle Inequality

We can rearrange the inequality from the similarity lemma to find

$$\sum_{a\in\mathcal{F}} J_s\left(a,c\right) - \sum_{a\in\mathcal{F}} J_s\left(a,b\right) \geq J_s\left(b,c\right) - 1$$

$$\sum_{a\in\mathcal{F}} \left(J_s\left(a,c\right) - J_s\left(a,b\right)\right) \geq J_s\left(b,c\right) - 1$$

$$\sum_{a\in\mathcal{F}} \left(\left(1 - d_s(a,c)\right) - \left(1 - d_s(a,b)\right)\right) \geq \left(1 - d_s(b,c)\right) - 1$$

$$\sum_{a\in\mathcal{F}} \left(d_s(a,b) - d_s(a,c)\right) \geq -d_s(b,c)$$

$$\sum_{a\in\mathcal{F}} \left(d_s(a,c) - d_s(a,b)\right) \leq d_s(b,c)$$

$$\sum_{a\in\mathcal{F}} d_s(a,c) \leq \sum_{a\in\mathcal{F}} d_s(a,b) + d_s(b,c)$$

From this rearranged form, we now complete the proof. For each finite $\mathcal{F} \subseteq \tilde{A}$ and each $b \in \tilde{B}, c \in \tilde{C}$, we have the inequality:

$$\sum_{a\in\mathcal{F}} d_s(a,c) \leq \sum_{a\in\mathcal{F}} d_s(a,b) + d_s(b,c)$$

Taking the supremum yields

$$\sup_{\mathcal{F}\subseteq\tilde{A},|\mathcal{F}|<\infty} \sum_{a\in\mathcal{F}} d_s(a,c) \leq \sup_{\mathcal{F}\subseteq\tilde{A},|\mathcal{F}|<\infty} \left(\sum_{a\in\mathcal{F}} d_s(a,b) + d_s(b,c)\right)$$

$$\sup_{\mathcal{F}\subseteq\tilde{A},|\mathcal{F}|<\infty} \sum_{a\in\mathcal{F}} d_s(a,c) \leq \sup_{\mathcal{F}\subseteq\tilde{A},|\mathcal{F}|<\infty} \left(\sum_{a\in\mathcal{F}} d_s(a,b)\right) + d_s(b,c)$$

$$\sum_{a\in\tilde{A}} d_s(a,c) \leq \sum_{a\in\tilde{A}} d_s(a,b) + d_s(b,c)$$

Now, for each $a \in \tilde{A}$, pick $b_a \in \tilde{B}$ such that

$$d_s(a,b_a) = \min_{b\in\tilde{B}} d_s(a,b)$$

and for each $b \in \tilde{B}$ pick $c_b \in \tilde{C}$ such that

$$d_s(b, c_b) = \min_{c \in \tilde{C}} d_s(b, c)$$

Now, fix $b \in \tilde{B}$. Let $I_b = \{a \in \tilde{A} | b_a = b\}$. Then applying the inequality with $\mathcal{F} = I_b$ and $c = c_b$ for each non-empty I_b yields:

$$\sum_{a \in I_b} d_s(a, c_b) \leq \sum_{a \in I_b} d_s(a, b) + d_s(b, c_b)$$

As

$$\min_{c \in \tilde{C}} d_s(a, c) \leq d_s(a, c_b)$$

and

$$d_s(b, c_b) = \min_{c \in \tilde{C}} d_s(b, c)$$

then

$$\sum_{a \in I_b} \min_{c \in \tilde{C}} d_s(a, c) \leq \sum_{a \in I_b} d_s(a, b) + \min_{c \in \tilde{C}} d_s(b, c)$$

Sum over $\tilde{B}$

$$\sum_{b \in \tilde{B}} \sum_{a \in I_b} \min_{c \in \tilde{C}} d_s(a, c) \leq \sum_{b \in \tilde{B}} \sum_{a \in I_b} d_s(a, b) + \sum_{b \in \tilde{B}} \min_{c \in \tilde{C}} d_s(b, c)$$

As $\{I_b | b \in \tilde{B}, I_b \neq \emptyset\}$ forms a partition of $\tilde{A}$, then

$$\sum_{a \in \tilde{A}} \min_{c \in \tilde{C}} d_s(a, c) \leq \sum_{a \in \tilde{A}} \min_{b \in \tilde{B}} d_s(a, b) + \sum_{b \in \tilde{B}} \min_{c \in \tilde{C}} d_s(b, c)$$

Thus

$$\overrightarrow{PCh}(\tilde{A}, \tilde{C}) \leq \overrightarrow{PCh}(\tilde{A}, \tilde{B}) + \overrightarrow{PCh}(\tilde{B}, \tilde{C})$$

Similarly, we can show

$$\overrightarrow{PCh}(\tilde{C}, \tilde{A}) \leq \overrightarrow{PCh}(\tilde{B}, \tilde{A}) + \overrightarrow{PCh}(\tilde{C}, \tilde{B})$$

Thus

$$\overrightarrow{PCh}(\tilde{A}, \tilde{C}) + \overrightarrow{PCh}(\tilde{C}, \tilde{A}) \leq \overrightarrow{PCh}(\tilde{A}, \tilde{B}) + \overrightarrow{PCh}(\tilde{B}, \tilde{C}) + \overrightarrow{PCh}(\tilde{B}, \tilde{A}) + \overrightarrow{PCh}(\tilde{C}, \tilde{B})$$

$$PCh(\tilde{A}, \tilde{C}) \leq PCh(\tilde{A}, \tilde{B}) + PCh(\tilde{B}, \tilde{C})$$

3.4 An Example Calculation

Consider the measure space Ω in Figure 2. As we are identifying partitions up to sets of measure 0, we can ignore measure 0 sets in the measure space, as shown in Figure 3.

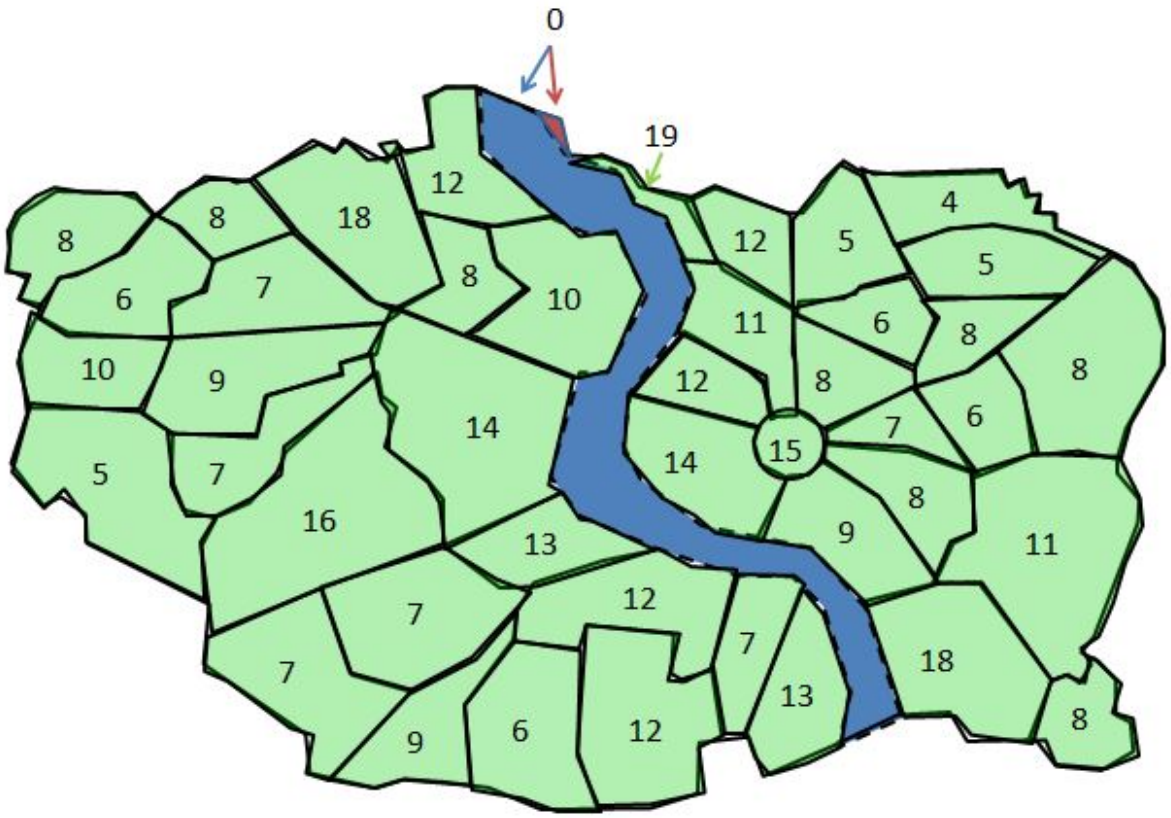


Figure 2: Consider the whole region in this image as Ω in our measure space. The measure of each green atom is given. The blue and red subsets are both of measure 0.

Two measurable partitions are given in Figure 4. We can find the part of B that is closest to A_1 under the Steinhaus metric as shown in Figure 5.

We repeat this process for all A_i and B_j to find $PCh(\tilde{A}, \tilde{B}) \approx 4.603$

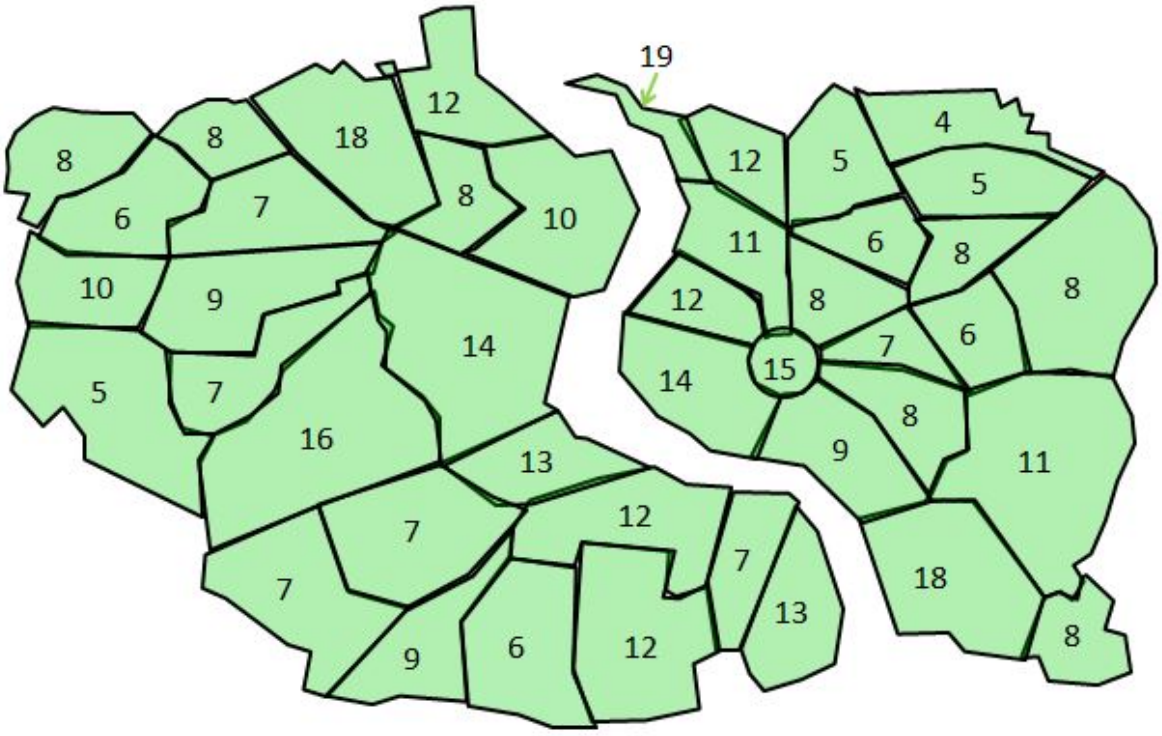
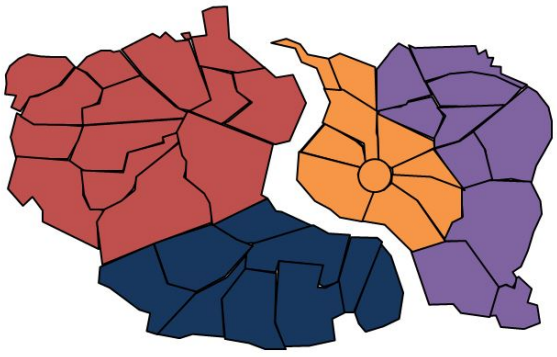
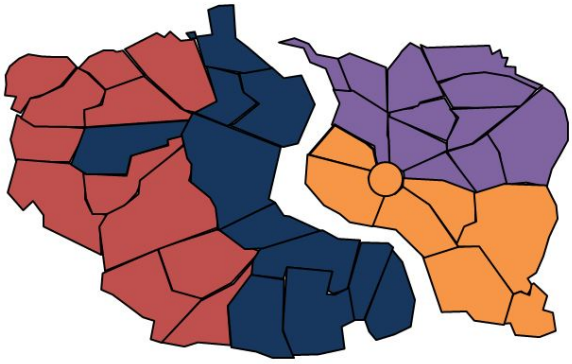


Figure 3: The subsets of measure 0 have been removed.

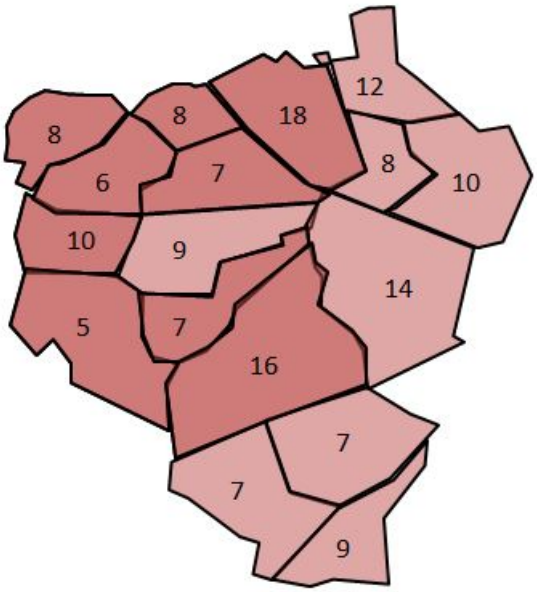


(a) Partition A. The parts of the partition are A_1, A_2, A_3, A_4 and are colored Red, Blue, Orange, and Purple respectively.

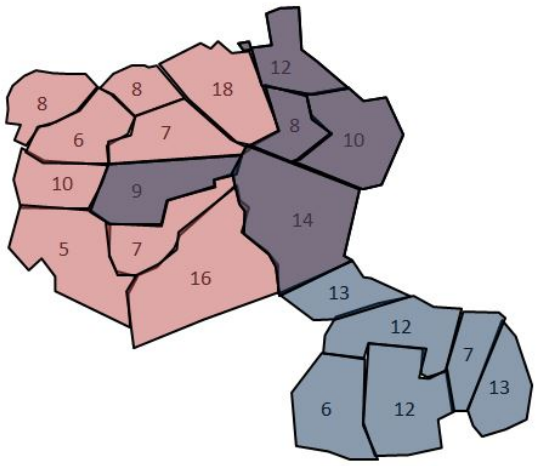


(b) Partition B. The parts of the partition are B_1, B_2, B_3, B_4 and are colored red, blue, orange, and purple respectively.

Figure 4: Two possible measurable partitions are given.



(a) Parts A_1 and B_1 are compared. We have $d_s(A_1, B_1) \approx 0.472$.



(b) Parts A_1 and B_2 are compared. We have $d_s(A_1, B_2) \approx 0.736$.

Figure 5: The two parts of partition B that intersect A_1 are shown. B_1 is the closest part of B to A_1. The remaining parts of B have a Steinhaus distance of 1 from A_1, since they do not intersect A_1.

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